

Option C: Neurobiology and Behaviour

9. Chacma baboons (*Papio ursinus*) are the largest members of the monkey family and are a highly social species that live in groups of up to 200 individuals. Within a group, adult males form a dominance hierarchy that is established and maintained by fighting and aggressive displays.



- (a) (i) State the meaning of the term ‘dominance hierarchy’. [1]

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- (ii) State and explain an advantage of dominance hierarchy in chacma baboons. [2]

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- (iii) Suggest an advantage of maintaining hierarchy by aggressive displays rather than fighting. [1]

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(iv) With reference to the appropriate branches of the nervous system, explain how the hypothalamus prepares the baboons for the ‘fight or flight response’ during these aggressive displays. [4]

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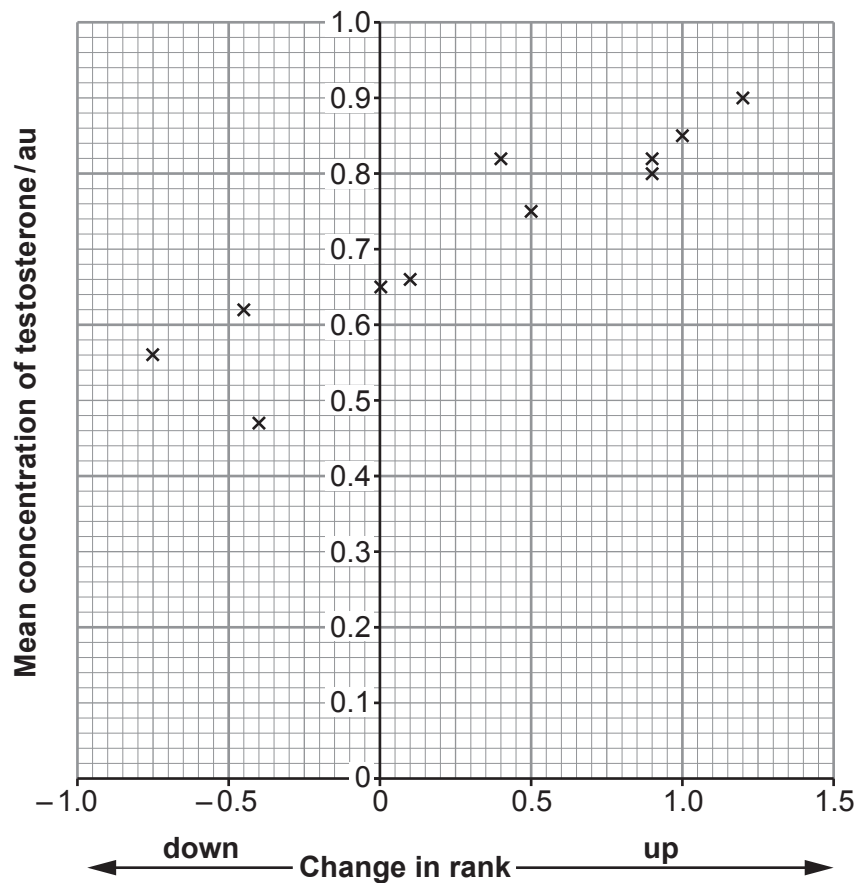
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- (b) Male ranking is unstable and changes every 6–12 months. This is because young males tend to migrate between groups and high-ranking males frequently lose their status to younger, immigrant males. Male testosterone levels were measured using faecal samples in a group of chacma baboons. **Graph 9.1** shows the relationship between change in male ranking and mean testosterone levels.

Graph 9.1



- (i) Calculate the percentage change in mean testosterone levels when the chacma baboon moves up one rank from 0 to 1. [2]

Percentage change =

- (ii) Suggest an advantage for the change in testosterone levels. [1]

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- (c) Courtship behaviour, as shown by the Indian peafowl (*Pavo cristatus*) in **Image 9.2**, is an example of innate behaviour.

Image 9.2

Female (peahen)



Male (peacock)



- (i) State the meaning of the term innate behaviour.

[1]

- (ii) Courtship behaviour, as seen in peafowl, is an example of the male handicap model. With reference to **Image 9.2**, state the evidence that this is an example of the male handicap model and conclude how courtship behaviour provides both an advantage and a disadvantage in this species.

[3]



- (d) Scientists carried out research into possible changes that occur in the brains of squirrels as a result of learning how to get nuts out of a bird feeder. MRI images can be analysed to measure the volume of the different parts of the hippocampus.



- (i) State **two** reasons why the scientists studied the hippocampus region of the brain [2]

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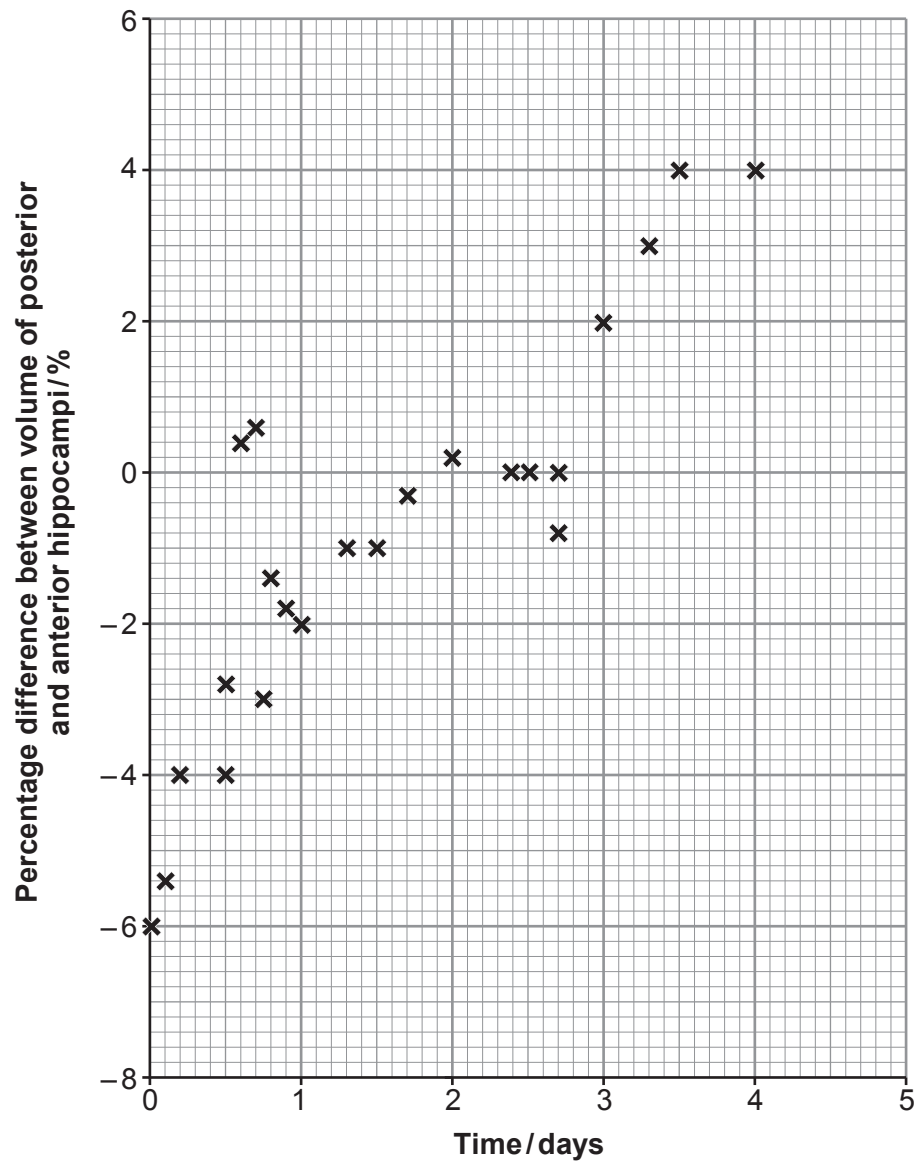
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Graph 9.3 shows the percentage differences between the volumes of the posterior and anterior hippocampi against the duration of time spent with the bird feeder.

Graph 9.3



(ii) State the trend shown in **Graph 9.3** and use your knowledge of brain structure and neuroplasticity to state **two** conclusions that can be drawn. [3]

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